

have no predictive value. This common pitfall of applying data mining to price data is the reason why the term often has derogatory connotations in reference to trading systems.

⁶To avoid hindsight bias error in developing trading systems, the available past data is segmented into seen data (i.e., “in-sample”) that is used for system development and unseen data (i.e., “out-of-sample”) that is used for system testing. Any results on the in-sample data are ignored because they are hindsight-biased. Although segmenting the data to reserve unseen data for testing is a necessary condition to avoid misleading results, it is not a sufficient condition as Woodruff goes on to explain.

⁷In 2011, QIM changed the exact calculation it used to reduce leverage during periods of poor performance, but the new formulation was similar in both conceptual and practical terms. It is, therefore, simpler to talk about their leverage reduction as one process.